

Consider this problem (sometimes called the longest run statistics [2]) about a random bit string: in a sequence of length N of independent events with the same probability distribution on a finite number of possible outcomes, what is the probability distribution of the length of the longest substring such that all events in the substring are of a specific outcome? The broken stick problem is: given a stick of length 1, if we break it at k random points, it will be broken into $k + 1$ pieces, and what is the probability distribution of the length of the longest piece? It turns out that the two problems are related: the continuum limit of the former problem is the generating function of the answer to the latter problem.

The formal statement of the problem is as follows: consider a random bit string of length $N \in \mathbb{N}$, where each bit is independent and has probability $p \in [0, 1]$ of being 1. Let $P_{N,S}(p)$ be the probability that the length of the longest all-1 substring of the bit string is $S \in \mathbb{N}$. What is the expression of $P_{N,S}(p)$? The recurrence relation is

$$P_{N,0 < S < N} = (1 - p) \left(p^S \sum_{s=0}^S P_{N-S-1,s} + \sum_{s=0}^{S-1} P_{N-s-1,S} \right),$$

with initial conditions

$$P_{N,0} = (1 - p)^N, \quad P_{N,N} = p^N, \quad P_{N,S > N} = 0.$$

Then, $P_{N,S}$ is a polynomial of degree N in p .

The continuum limit is defined as

$$f(P, x) := \lim_{N \rightarrow \infty} N P_{N,xN} \left(P^{1/N} \right), \quad P \in [0, 1], \quad x \geq 0,$$

which is a probability distribution of $x := S/N$ with $P := p^N$ as the parameter. Its exact expression is

$$f(P = 0, x) = \delta(x), \quad f(P \in (0, 1], x) = \begin{cases} 0, & x = 0, \\ \text{complicated}, & x \in (0, 1), \\ P\delta(x - 1), & x \geq 1. \end{cases}$$

where

$$\begin{aligned} \text{complicated} = & -P^x \ln P (2 + \ln P^{x-1}) + \ln P \sum_{k=2}^{\lceil 1/x \rceil - 1} \frac{k(-1)^k}{(k-2)!} P^{kx} (\ln P^{kx-1})^{k-2} \\ & \times \left(1 + \frac{2}{k-1} \ln P^{kx-1} + \frac{1}{k(k-1)} (\ln P^{kx-1})^2 \right). \end{aligned}$$

On the other hand, the continuum limit is also the probability distribution of the longest interval without occurrences in the Poisson process on the interval $[0, 1]$ with rate $-\ln P$. When there are k occurrences each at positions $x_1 < \dots < x_k$, it divides the interval into $k + 1$ intervals:

$$S_0 = x_1, \quad S_1 = x_2 - x_1, \quad \dots, \quad S_k = 1 - x_k.$$

Order them into $S_{(0)} \leq \dots \leq S_{(k)}$. The probability distribution of the longest interval is then $f(x, P)$.

If the number k is not Poisson distributed but fixed instead, the problem is called the broken stick problem, and the probability distribution of $S_{(k)}$ is [1]

$$f_{k>0}(x) = \sum_{j=0}^{k+1} -jk \binom{k+1}{j} (-1)^j \max(1 - jx, 0)^{k-1}, \quad f_0(x) = \delta(x - 1).$$

The answers to the two problems are related:

$$f(x, P) = \sum_{k=0}^{\infty} \frac{P(-\ln P)^k}{k!} f_k(x).$$

Therefore, we can see that $f(x, P)/P$ as a function of $-\ln P$ is the exponential generating function of $f_k(x)$.

- [1] Lars Holst. “On the lengths of the pieces of a stick broken at random”. en. In: *Journal of Applied Probability* 17.3 (Sept. 1980), pp. 623–634. ISSN: 0021-9002, 1475-6072. DOI: 10.2307/3212956.
- [2] Frosso S. Makri, Andreas N. Philippou, and Zaharias M. Psillakis. “Shortest and longest length of success runs in binary sequences”. en. In: *Journal of Statistical Planning and Inference* 137.7 (July 2007), pp. 2226–2239. ISSN: 03783758. DOI: 10.1016/j.jspi.2006.07.007.